

Jesse J. Hughes

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github.com/scallyw4g

12 Years Professional Experience, 10 Years Remote



Windows
Linux
C
Metaprogramming
Shell
Git



C++
OpenGL & WebGL
SIMD
Multithreading



Web Development
Ruby / Rails
Javascript / Typescript
Golang

Interests

Developer Tooling
Performance
Graphics & Game Engines

OSS Contributions

Poof

<https://scallyw4g.github.io/poof/>

Fast and flexible metaprogramming system

Brings features of modern programming languages to C

Metaprogramming Compiler

2020 - Present

C C++

Bonsai

<https://github.com/scallyw4g/bonsai>

Renderer, Profiler, Entity System, Physics

From scratch -- no third party libraries

3D Graphics & Game Engine

2015 - Present

OpenGL C C++

Oil

<https://github.com/oilshell/oil>

Upgrade path from bash to a better system scripting language and runtime

Contributed a (partial) Python runtime and a mark/sweep garbage collector

POSIX Shell & Python Runtime

2022

C++ Python Shell

Allspice

Lead Rendering Engineer

Prototype and implement 2D renderer for web-based EE CAD tool

2023

Handle up to low billions of elements in realtime, on midrange hardware

Rust C++ WebGL WASM JS/TS React

Integrate renderer into existing React product

Remote

Lemurian Labs

Compiler & Cycle-Simulator Lead

Lead development of greenfield compiler, runtime and simulation software projects

2022 - 2023

Hire, manage, train engineers ranging in skill from Intern to Principal

C C++ Multithreading golang

Consistently delivered high quality results on-time, often with scarce resources

Remote

Optic Power

Solutions Architect & Reverse Engineer

Own discovery, architecture and kickoff phases of client projects

2020 - 2022

Manage up to five in-flight projects, including the technical staff, sales team and client

TS JS Elixir C C++ ASM

Reverse engineer closed-source binaries, including Valves Source2 engine -- "Crazy game hacks" was in the job description

Remote

KeyDB

Senior Engineer

Performance-oriented contract work on a multithreaded Redis fork

2021

Rewrote IO subsystems to use io_uring

C C++ Multithreading

Remote

Evolution Ventures

Graphics Engineer

Build, extend, maintain WebGL-based architecture visualization tools

2019 - 2021

High-level technical direction for 3D products

WebGL TS React

Remote

Web Development & Consulting

CrossFit Games -- games.crossfit.com

McAfee MPOWER -- mcafeempower.com

Many other clients including Stanford and Gap

Fullstack Developer

2013 - 2019

JS PHP

Remote & Onsite

Fresche Solutions

Rebuilt product integration portal - C++ backend, Angular2 frontend

Refactored 300Kloc C++ Windows IDE to support Unicode

Increased test coverage by 4x

Win32 Engineer

2014 - 2016

C++ JS PHP

Onsite